화학수학

열네 번째 수업

미분방정식 / 연산자

Inhomogeneous Linear ODE

$$a\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + b\frac{\mathrm{d}y}{\mathrm{d}x} + cy = g(x)$$

1. Solve the complementary equation → complementary function (보상함수) y_c
2. Find a particular solution y_p that satisfies the DE
3. The general solution is $y = y_\mathrm{c} + y_\mathrm{p}$

Guess for Particular Solution

Inhomogeneous term	Trial solution
1	A
t^n	$A_0 + A_1 t + A_2 t + \cdots + A_n t^n$
$e^{\alpha t}$	$Ae^{\alpha t}$
$t^n e^{\alpha t}$	$e^{\alpha t}(A_0 + A_1 t + A_2 t + \cdots + A_n t^n)$
$e^{\alpha t} \sin(\beta t)$	$e^{\alpha t}[A \cos(\beta t) + B \sin(\beta t)]$
$e^{\alpha t} \cos(\beta t)$	$e^{\alpha t}[A \cos(\beta t) + B \sin(\beta t)]$

(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Table 12.1)

Glitch in the Method

A trial solution y_p can be

a solution of the complementary equation.

In this case, try $x^n y_\mathrm{p}$ instead!

(n: smallest integer that resolves the problem)

Application: Forced Harmonic Oscillator

$$F = m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -kx + \underbrace{F(t)}_{\text{external force}}$$

- Standard form

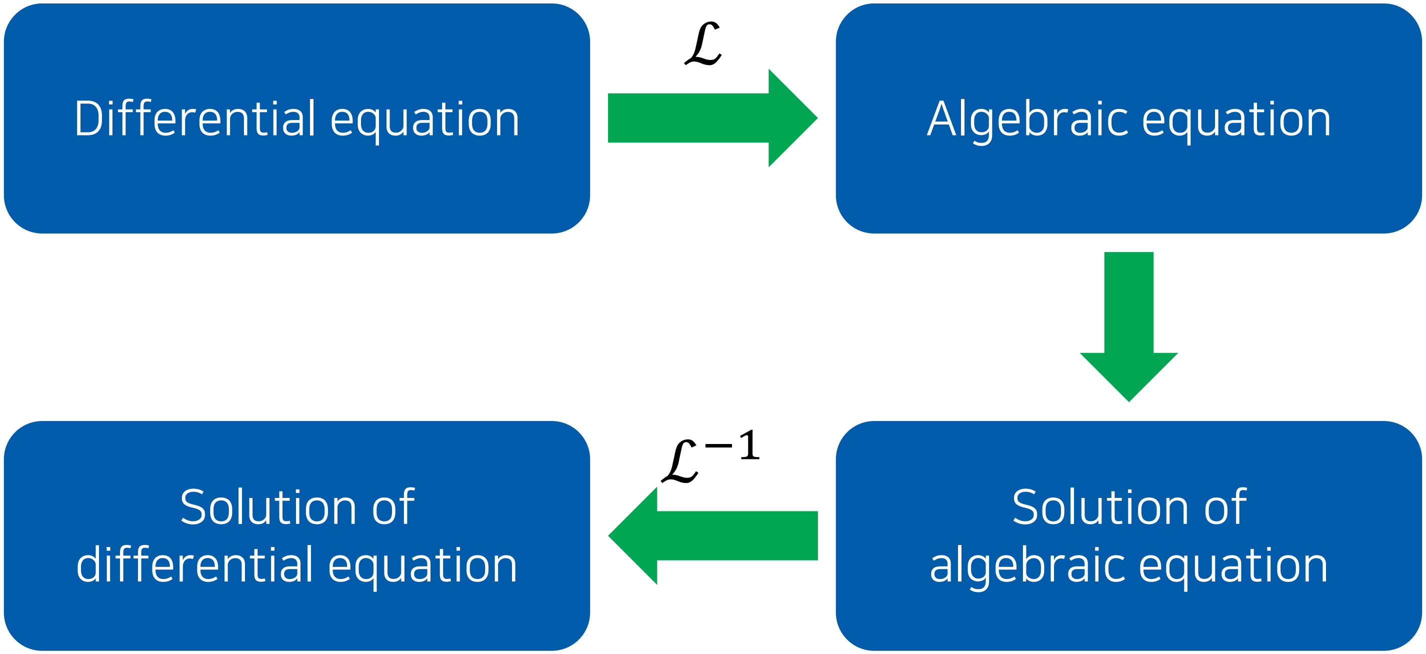
$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{k}{m}x = \frac{F(t)}{m}$$

- Complementary equation

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{k}{m}x = 0$$

$$\to x_\mathrm{c}(t) = c_1\cos\omega t + c_2\sin\omega t$$

Solving DE with Laplace Transforms



Today's Class

- Partial differential equations
  - Separation of variables
  - Wave equation
  - Schrödinger equation
- Mathematical operators
  - Operators
  - Operator algebra

Partial Differential Equations

- General form of a linear second-order PDE

$$A\frac{\partial^2 u}{\partial x^2} + B\frac{\partial^2 u}{\partial x \partial y} + C\frac{\partial^2 u}{\partial y^2} + D\frac{\partial u}{\partial x} + E\frac{\partial u}{\partial y} + Fu = G$$

where $A, B, \cdots, G$ are functions of x and y.

- $G(x, y) = 0$: homogeneous
- $G(x, y) \neq 0$: inhomogeneous

Separation of Variables

$$A\frac{\partial^2 u}{\partial x^2} + B\frac{\partial^2 u}{\partial x \partial y} + C\frac{\partial^2 u}{\partial y^2} + D\frac{\partial u}{\partial x} + E\frac{\partial u}{\partial y} + Fu = G$$

$X' = \mathrm{d}X/\mathrm{d}x\,, X'' = \mathrm{d}^2X/\mathrm{d}x^2, \cdots$
$Y' = \mathrm{d}Y/\mathrm{d}y\,, Y'' = \mathrm{d}^2Y/\mathrm{d}y^2, \cdots$

Assume that $u(x, y) = X(x)Y(y)$.

If we obtain $f(x, X, X', \cdots) = g(y, Y, Y', \cdots)$, let

$$f(x, X, X', \cdots) = g(y, Y, Y', \cdots) = \lambda$$

where λ is the separation constant.

(Tip: it is often convenient to use $\lambda = \kappa^2$ or $-\kappa^2$)

Not in Mortimer

Example

Find solutions of

$$\frac{\partial^2 u}{\partial x^2} = 4\frac{\partial u}{\partial y}.$$

Assume $u(x, y) = X(x)Y(y)$ and obtain

$$X''Y = 4XY'$$

$$\frac{X''}{4X} = \frac{Y'}{Y}$$

Each side must be a constant. Consider the following three cases:

1) Separation constant > 0
2) Separation constant < 0
3) Separation constant = 0

Example

Find solutions of

$$\frac{\partial^2 u}{\partial x^2} = 4\frac{\partial u}{\partial y}.$$

1) Case 1:

$$\frac{X''}{4X} = \frac{Y'}{Y} = \kappa^2$$

Hence,

$$X'' - 4\kappa^2 X = 0,\ Y' - \kappa^2 Y = 0$$

which have the solutions

$$X(x) = c_1 e^{2\kappa x} + c_2 e^{-2\kappa x},\ Y(y) = c_3 e^{\kappa^2 y}$$

Not in Mortimer

Example

Find solutions of

$$\frac{\partial^2 u}{\partial x^2} = 4\frac{\partial u}{\partial y}.$$

1) Case 1:

$$X(x) = c_1 e^{2\kappa x} + c_2 e^{-2\kappa x},\ Y(y) = c_3 e^{\kappa^2 y}$$

The final solution is

$$u(x,y) = X(x)Y(y) = (c_1 e^{2\kappa x} + c_2 e^{-2\kappa x})c_3 e^{\kappa^2 y}$$

or

$$u(x,y) = A_1 e^{2\kappa x+\kappa^2 y} + B_1 e^{-2\kappa x+\kappa^2 y}$$

Not in Mortimer

Example

Find solutions of

$$\frac{\partial^2 u}{\partial x^2} = 4\frac{\partial u}{\partial y}.$$

2) Case 2:

$$\frac{X''}{4X} = \frac{Y'}{Y} = -\kappa^2$$

Hence,

$$X'' + 4\kappa^2 X = 0,\ Y' + \kappa^2 Y = 0$$

which have the solutions

$$X(x) = c_4 \cos 2\kappa x + c_5 \sin 2\kappa x\,,\ Y(y) = c_6 e^{-\kappa^2 y}$$

Not in Mortimer

Example

Find solutions of

$$\frac{\partial^2 u}{\partial x^2} = 4\frac{\partial u}{\partial y}.$$

2) Case 2:

$$X(x) = c_4 \cos 2\kappa x + c_5 \sin 2\kappa x\,,\ \ Y(y) = c_6 e^{-\kappa^2 y}$$

The final solution is

$$u(x,y) = X(x)Y(y) = (c_4 \cos 2\kappa x + c_5 \sin 2\kappa x)c_6 e^{-\kappa^2 y}$$

or

$$u(x,y) = A_2 e^{-\kappa^2 y}\cos 2\kappa x + B_2 e^{-\kappa^2 y}\sin 2\kappa x$$

Not in Mortimer

Example

Find solutions of

$$\frac{\partial^2 u}{\partial x^2} = 4\frac{\partial u}{\partial y}.$$

3) Case 3:

$$\frac{X''}{4X} = \frac{Y'}{Y} = 0$$

Hence,

$$X'' = Y' = 0$$

which have the solutions

$$X(x) = c_7 x + c_8,\ Y(y) = c_9$$

Example

Find solutions of

$$\frac{\partial^2 u}{\partial x^2} = 4\frac{\partial u}{\partial y}.$$

3) Case 3:

$$X(x) = c_7 x + c_8, \;\; Y(y) = c_9$$

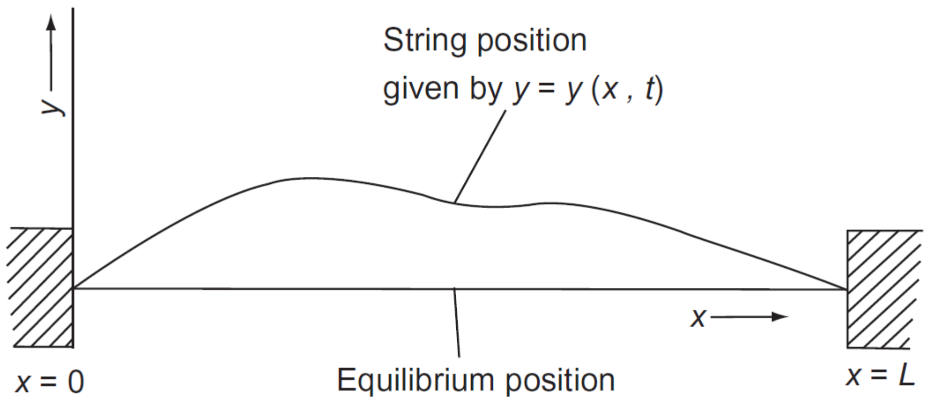
The final solution is

$$u(x, y) = X(x)Y(y) = (c_7 x + c_8)c_9$$

or

$$u(x, y) = A_3 x + B_3$$

Waves in a String



- Assumptions
  1. The string is perfectly flexible and homogeneous
  2. The displacements y are small compared to L
  3. It has a finite length and both ends are fixed

(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 12.6)

Wave Equation (파동방정식)

From the equation of motion (Newton's law),

$$\frac{\partial^2 y}{\partial t^2} = \frac{T}{\rho}\frac{\partial^2 y}{\partial x^2}$$

where T is the magnitude of the tension force and ρ is the mass of the string per unit length.

Letting $c = \sqrt{T/\rho}$,

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$$

Additional Conditions

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$$

- Boundary conditions

$$y(0, t) = y(L, t) = 0$$

- Initial conditions

$$y(x, 0) = 0, \qquad \left.\frac{\partial y}{\partial t}\right|_{t=0} = f(x)$$

Solving Wave Equation

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$$

Assume $y(x,t) = \psi(x)\theta(t)$;

$$\psi \frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} = c^2 \frac{\mathrm{d}^2\psi}{\mathrm{d}x^2}\theta$$

$$\underbrace{\frac{1}{c^2\theta}\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2}}_{\text{temporal part}} = \underbrace{\frac{1}{\psi}\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2}}_{\text{spatial part}} = \lambda$$

Spatial Part

$$\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} = \lambda\psi$$

1. If $\lambda > 0$, we can set $\lambda = \kappa^2$ and obtain

$$\psi(x) = a_1 e^{\kappa x} + a_2 e^{-\kappa x}$$

2. If $\lambda = 0$, we obtain

$$\psi(x) = a_3 x + a_4$$

3. If $\lambda < 0$, we can set $\lambda = -\kappa^2$ and obtain

$$\psi(x) = a_5 \cos \kappa x + a_6 \sin \kappa x$$

Applying Boundary Conditions

$$y(0,t) = y(L,t) = 0 \ \rightarrow \ \psi(0) = \psi(L) = 0$$

1. If $\lambda > 0$, $\psi(x) = a_1 e^{\kappa x} + a_2 e^{-\kappa x}$

$$\psi(0) = a_1 + a_2 = 0, \ \psi(L) = a_1 e^{\kappa L} + a_2 e^{-\kappa L} = 0$$

$$\therefore a_1 = a_2 = 0$$

2. If $\lambda = 0$, $\psi(x) = a_3 x + a_4$

$$\psi(0) = a_4 = 0, \ \psi(L) = a_3 L + a_4 = 0$$

$$\therefore a_3 = a_4 = 0$$

Applying Boundary Conditions

$$y(0,t) = y(L,t) = 0 \;\rightarrow\; \psi(0) = \psi(L) = 0$$

3. If $\lambda < 0$, $\psi(x) = a_5 \cos \kappa x + a_6 \sin \kappa x$

$$\psi(0) = a_5 = 0,\;\; \psi(L) = a_5 \cos \kappa L + a_6 \sin \kappa L = 0$$

$\therefore a_5 = 0$ and $\kappa = n\pi/L$

The only non-zero solution is

$$\psi(x) = a_6 \sin\left(\frac{n\pi x}{L}\right)$$

when $\lambda = -\kappa^2$ and $\kappa = n\pi/L$

Temporal Part

$$\frac{1}{c^2\theta}\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} = \frac{1}{\psi}\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} = \lambda = -\kappa^2 = -\left(\frac{n\pi}{L}\right)^2$$

$$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} = -\left(\frac{n\pi}{L}\right)^2 c^2\theta$$

Since the separation constant is negative,

$$\theta(t) = b_1\cos\left(\frac{n\pi c}{L}t\right) + b_2\sin\left(\frac{n\pi c}{L}t\right)$$

Applying Initial Conditions

$$\theta(t) = b_1 \cos\left(\frac{n\pi c}{L} t\right) + b_2 \sin\left(\frac{n\pi c}{L} t\right)$$

$$y(x, 0) = 0 \rightarrow \theta(0) = 0$$

Substitution leads to

$$\theta(0) = b_1 = 0$$

Hence,

$$\theta(t) = b_2 \sin\left(\frac{n\pi c}{L} t\right)$$

Total Solution

- Spatial solution

$$\psi(x) = a_6 \sin\left(\frac{n\pi x}{L}\right)$$

- Temporal solution

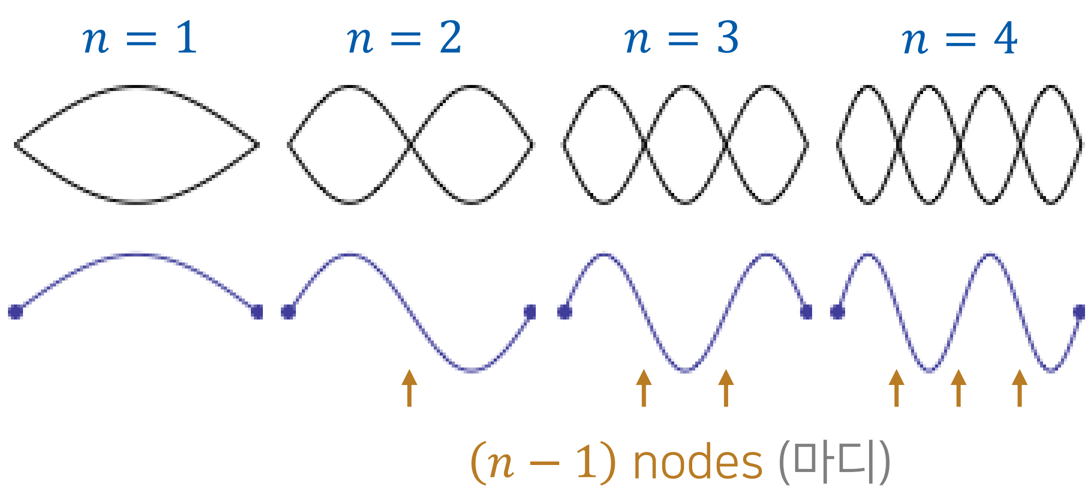
$$\theta(t) = b_2 \sin\left(\frac{n\pi c t}{L}\right)$$

- Total solution

$$y(x,t) = \psi(x)\theta(t) = A \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{n\pi c t}{L}\right)$$

Standing Waves (정상파)

$$y(x,t) = \psi(x)\theta(t) = A\sin\left(\frac{n\pi x}{L}\right)\sin\left(\frac{n\pi ct}{L}\right)$$

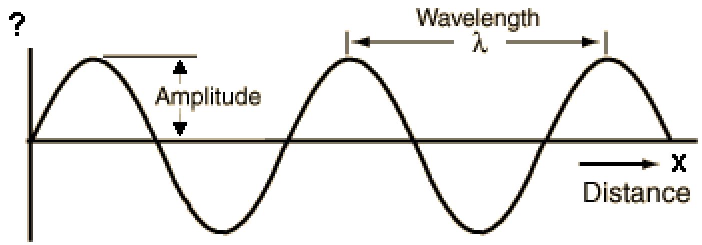


(Image: https://www.acs.psu.edu/drussell/demos/standingwaves/standingwaves.html)

§ 12.7.2

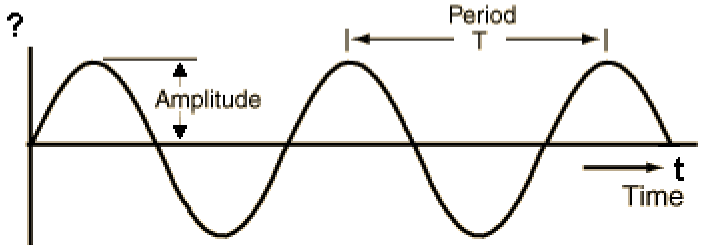
Wavelength and Period (파장과 주기)

$$y(x,t) = \psi(x)\theta(t) = A\sin\left(\frac{n\pi x}{L}\right)\sin\left(\frac{n\pi ct}{L}\right)$$



- Wavelength λ

$$n\pi\lambda/L = 2\pi \rightarrow \lambda = 2L/n$$



- Period τ

$$n\pi c\tau/L = 2\pi \rightarrow \tau = 2L/nc$$

(Image: https://www.everythingrf.com/rf-calculators/frequency-to-wavelength)

Traveling Waves (진행파)

$$y(x,t) = A\sin[k(x-ct)]$$

(Image: https://spacelights.tumblr.com/post/37429141540/infinity-imagined-standing-waves-a-standing-wave)

Example 12.12

Show that the function $y(x,t) = A\sin[k(x-ct)]$ satisfies the wave equation

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}.$$

The left-handed side is

$$\frac{\partial^2 y}{\partial t^2} = -Ak^2c^2\sin[k(x-ct)]$$

and the right-handed side is

$$c^2\frac{\partial^2 y}{\partial x^2} = -Ak^2c^2\sin[k(x-ct)]$$

Hence, the function satisfies the wave equation.

The Schrödinger Equation

- 1-dimensional time-dependent Schrödinger equation

$$i\hbar\frac{\partial\Psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\Psi}{\partial x^2} + \mathcal{V}(x,t)\Psi$$

  - $\Psi = \Psi(x,t)$: wave function
  - m: mass of a particle
  - $\mathcal{V}(x,t)$: potential energy
  - $\hbar = h/2\pi$, where h is Planck's constant

- Consider the case $\mathcal{V}(x,t) = \mathcal{V}(x)$!

Separation of Variables

$$i\hbar\frac{\partial\Psi}{\partial t}=-\frac{\hbar^2}{2m}\frac{\partial^2\Psi}{\partial x^2}+\mathcal{V}(x)\Psi$$

Assume $\Psi(x,t)=\psi(x)\theta(t)$:

$$i\hbar\psi(x)\theta'(t)=-\frac{\hbar^2}{2m}\psi''(x)\theta(t)+\mathcal{V}(x)\psi(x)\theta(t)$$

Dividing both sides by $\psi(x)\theta(t)$,

$$\underbrace{i\hbar\frac{\theta'(t)}{\theta(t)}}_{\text{depends only on } t}=\underbrace{-\frac{\hbar^2}{2m}\frac{\psi''(x)}{\psi(x)}+\mathcal{V}(x)}_{\text{depends only on } x}$$

Separation Constant

$$i\hbar\frac{\theta'(t)}{\theta(t)} = -\frac{\hbar^2}{2m}\frac{\psi''(x)}{\psi(x)} + \mathcal{V}(x) = E$$

- Spatial part (time-independent Schrödinger eqn.)

$$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} + \mathcal{V}(x)\psi(x) = E\psi(x)$$

- Temporal part

$$i\hbar\frac{\mathrm{d}\theta}{\mathrm{d}t} = E\theta(t)$$

$$\theta(t) = e^{-iEt/\hbar}$$

13. Operators, Matrices, and Group Theory

(13장 연산자, 행렬, 군론)

Mathematical Operators (연산자)

perform mathematical operations on some function

e.g. "+3" operator, "×2" operator, "$\mathrm{d}/\mathrm{d}x$" operator

- Typically denoted by a letter with a caret ($\hat{}$)
- Operate on the function on its right

Some Examples

- Multiplication operators multiply a function by a constant or by another function
- Derivative operators differentiate a function one or more times
- Symmetry operators can move a point in space but can also operate on functions

Example 13.1

For the operator

$$\hat{A} = x + \frac{\mathrm{d}}{\mathrm{d}x},$$

find $\hat{A}f$ if $f = a \sin bx$, where a and b are constants.

Substitution gives

$$\hat{A}f = \left(x + \frac{d}{dx}\right)(a \sin bx) = ax \sin bx + ab \cos bx$$

Special Operators

- Identity operator (항등 연산자) $\hat{E}$: multiplication by 1 (doing nothing)

$$\hat{E}f = f$$

- Null operator (영 연산자) $\hat{0}$: multiplication by 0

$$\hat{0}f = 0$$

Operator Algebra

- Sum of two operators: if $\hat{C} = \hat{A} + \hat{B}$, then

$$\hat{C}f = (\hat{A} + \hat{B})f = \hat{A}f + \hat{B}f$$

- Product of two operators: if $\hat{C} = \hat{A}\hat{B}$, then

$$\hat{C}f = \hat{A}(\hat{B}f)$$

Example 13.3

Find the operator equal to the operator product

$$\hat{A} = \frac{\mathrm{d}}{\mathrm{d}x} x.$$

Take an arbitrary function $f = f(x)$;

$$\hat{A}f = \frac{\mathrm{d}}{\mathrm{d}x}(xf) = f + x\frac{\mathrm{d}f}{\mathrm{d}x} = \left(\hat{E} + x\frac{\mathrm{d}}{\mathrm{d}x}\right)f$$

Hence, the operator $\hat{A}$ is equal to

$$\hat{E} + x\frac{\mathrm{d}}{\mathrm{d}x}.$$

Properties of Operator Multiplication

- Associative property (결합법칙)

$$(\hat{A}\hat{B})\hat{C} = \hat{A}(\hat{B}\hat{C})$$

- Distributive property (분배법칙)

$$\hat{A}(\hat{B} + \hat{C}) = \hat{A}\hat{B} + \hat{A}\hat{C}$$

- Commutative property (교환법칙) → Not necessarily

$$\hat{A}\hat{B} = \hat{B}\hat{A}\,?$$

More on Commutative Property

- If $\hat{A}\hat{B} = \hat{B}\hat{A}$, $\hat{A}$ and $\hat{B}$ are said to commute

- Commutator (교환자)

$$[\hat{A},\hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

- If $\hat{A}$ and $\hat{B}$ commute,

$$[\hat{A},\hat{B}] = \hat{0}$$

Example 13.4

Find the commutator

$$\left[\frac{\mathrm{d}}{\mathrm{d}x}, x\right].$$

Take an arbitrary function $f = f(x)$;

$$\left[\frac{\mathrm{d}}{\mathrm{d}x}, x\right] f = \left(\frac{\mathrm{d}}{\mathrm{d}x} x - x \frac{\mathrm{d}}{\mathrm{d}x}\right) f = \frac{\mathrm{d}}{\mathrm{d}x}(xf) - x \frac{\mathrm{d}f}{\mathrm{d}x}$$

$$= f + x \frac{\mathrm{d}f}{\mathrm{d}x} - x \frac{\mathrm{d}f}{\mathrm{d}x} = f = \hat{E} f$$

Therefore,

$$\left[\frac{\mathrm{d}}{\mathrm{d}x}, x\right] = \hat{E}.$$

Rules for Commutativity

1. Operators acting on different independent variables commute

$$[x, \mathrm{d}/\mathrm{d}y] = 0$$

2. Multiplication and derivative operators usually do not commute

3. Two multiplication operators commute

4. Operator for multiplication by a constant commutes with any other operator

Exponentiation of Operator

(연산자의 거듭제곱)

$$\hat{A}^n = \hat{A}\hat{A}\cdots\hat{A} \ (n \text{ times})$$

Example 13.5

If $\hat{A} = x + \mathrm{d}/\mathrm{d}x$, find $\hat{A}^2$.

$$\hat{A}^2 = \hat{A}\hat{A} = \left(x + \frac{\mathrm{d}}{\mathrm{d}x}\right)\left(x + \frac{\mathrm{d}}{\mathrm{d}x}\right)$$

$$= x^2 + \frac{\mathrm{d}}{\mathrm{d}x}x + x\frac{\mathrm{d}}{\mathrm{d}x} + \frac{\mathrm{d}^2}{\mathrm{d}x^2}$$

Example 13.6

For the operator $\hat{A} = x + \mathrm{d}/\mathrm{d}x$, find $\hat{A}^2 f$ if $f(x) = \sin ax$.

From Example 13.5,

$$\hat{A}^2 = x^2 + \frac{\mathrm{d}}{\mathrm{d}x}x + x\frac{\mathrm{d}}{\mathrm{d}x} + \frac{\mathrm{d}^2}{\mathrm{d}x^2}$$

Substitution leads to

$$\hat{A}^2 f = \left(x^2 + \frac{\mathrm{d}}{\mathrm{d}x}x + x\frac{\mathrm{d}}{\mathrm{d}x} + \frac{\mathrm{d}^2}{\mathrm{d}x^2}\right)(\sin ax)$$

$$\hat{A}^2 f = x^2 \sin ax + \frac{\mathrm{d}}{\mathrm{d}x}(x \sin ax) + x\frac{\mathrm{d}}{\mathrm{d}x}(\sin ax) + \frac{\mathrm{d}^2}{\mathrm{d}x^2}(\sin ax)$$

$$= x^2 \sin ax + \sin ax + ax \cos ax + ax \cos ax - a^2 \sin ax$$

$$\hat{A}^2 f = x^2 \sin ax + \sin ax + 2ax \cos ax - a^2 \sin ax$$

Inverse of Operator (역 연산자)

- $\hat{A}^{-1}$: inverse of operator $\hat{A}$

$$\hat{A}\hat{A}^{-1} = \hat{A}^{-1}\hat{A} = \hat{E}$$

- $\hat{A}$ is also the inverse of $\hat{A}^{-1}$

- Not all operators have inverses (*e.g.* $\hat{0}$)